

RESEARCH ARTICLE

RR-HCL-SVM: A two-stage framework for assessing remaining thyroid tissue post-thyroidectomy in SPECT images

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Abstract

This paper presents a two-stage deep learning framework, RR-HCL-SVM, designed to aid in the assessment of residual thyroid tissues following thyroidectomy, utilizing single-photon emission computed tomography (SPECT) images. Leveraging the power of deep learning, our model offers a comprehensive solution for the detection and assessment of remaining thyroid tissues. To enhance accuracy, we introduce a unique combination of features, incorporating the Radio Scan Index (RSI) and radiomics features. These features not only improve accuracy but also provide valuable insights into tissue characterization. Moreover, we employ efficient clustering techniques for feature dimension reduction, preserving model performance while reducing computational complexity. Experimental results demonstrate the effectiveness of our approach, achieving an impressive F1 score of 0.97, sensitivity of 0.96, and specificity of 0.98. The RR-HCL SVM framework holds great promise in the clinical setting for the precise evaluation of residual thyroid tissues post-thyroidectomy, offering potential benefits for patient care and treatment planning.

KEYWORDS

clustering, deep learning, feature combination, machine learning, medical imaging, radiomics, residual thyroid tissues, SPECT images, thyroidectomy

1 | INTRODUCTION

Thyroid cancer is often the primary malignancy in cases of endocrine diseases. In Vietnam, thyroid cancer ranks fifth in frequency of occurrence among cancers, particularly in women, as per Globocan 2018 statistics. Differentiated thyroid cancer (DTC) accounts for 97%

of which includes papillary, follicular, and Hurthle cells.

The diagnostic evaluation after the surgery involves checking the presence of any residual thyroid tissue in the patient's body, and administering a therapeutic dose of ¹³¹I to eliminate the tissue if needed. To assess the patient's condition, the doctor uses clinical markers such

as TG, TSH, and single-photon emission computed tomography (SPECT) scintigraphy with a dose of 2–5 mCi of ^{131}I .

In the present era of relentless advancements in artificial intelligence (AI), the utilization of AI as a prospective method for assisting in the diagnosis of the presence or absence of residual thyroid tissue post-surgery and throughout the treatment process has gained substantial credence. While research applications of machine learning and deep learning in thyroid cancer are widespread, particularly in SPECT images, studies evaluating residual shielding post-surgery remain limited. Different from ultrasound images, the SPECT image is prone to noise from various factors including equipment condition, patient's physical condition, and time elapsed after using ^{131}I , thus requiring us proposed proper model and data preprocessing pipeline. In this study, we propose a two-stage framework for evaluating residual shielding with a suitable data preprocessing pipeline, aiming to enhance the accuracy compared to the previous study.¹ The main contribution of this work includes the following:

1. *Two-Stage Deep Learning Model*: We introduce a two-stage deep learning model for the precise detection and assessment of residual thyroid tissues. This model offers an advanced and effective approach for the medical community, addressing a critical need in post-thyroidectomy assessment.
2. *Feature Combination*: By incorporating a fusion of Radio Scan Index (RSI) and radiomics features, the framework achieves a remarkable enhancement in accuracy. This contribution underscores the importance of leveraging diverse data sources for improved diagnostic capabilities.
3. *Feature Dimension Reduction with Clustering*: The integration of clustering techniques for feature dimension reduction proves to be a valuable strategy for preserving model performance. This approach not only streamlines the feature space but also retains essential information, contributing to the model's robustness and efficiency.
4. *Outstanding Diagnostic Performance*: The proposed RR-HCL-SVM framework achieves outstanding results, as evidenced by an impressive F1 score of 0.97, sensitivity of 0.96, and specificity of 0.98. These exceptional performance metrics demonstrate the practical utility of the framework in clinical settings, offering a reliable and accurate method for assessing residual thyroid tissues.

The rest of the paper is organized as follows. Section 2 reviews related research on residual thyroid tissue assessment. In Section 3, we explore data set as

well as analyze possible challenges for model's learning. Section 4 describes our method for thyroid residual diagnosis and detection in detail. The experimental settings and results are presented and discussed in Section 6. Finally, we summarize our contribution to this research and the new enhanced direction in the future in Sections 7 and 8.

2 | RELATED WORKS

Currently, in addition to the traditional methods employed by physicians for diagnosis based on experience and SPECT imaging, there is a growing emergence of artificial intelligence (AI) applications in the diagnostic process, especially in post-operative residual assessment.

Firstly, the introduction of the Radioiodine Scan Index (RSI) has contributed to quantifying the diagnostic process for physicians using imaging.² The study acknowledges that there are many limitations. First, it is a retrospective analysis with a small sample size; therefore, any conclusions may be limited in their implications. Second, although the serum TSH level was well controlled, variation still remains in the serum Tg level influenced by the TSH level beyond the measurement limit (>80 mU/L).^{3–5} Third, circulating the ATA may cause false-negative or -positive results in the serum Tg level. Fourth, in WBS, the scanning time and treatment dose, which can cause variation in the radioiodine uptake, was not fully controlled. The last limitation is that neither the serum Tg level nor radioiodine WBS can accurately predict the 'true' metastatic tumor burden. The quantification of medical diagnoses made by physicians through image analysis has laid a foundation for the integration of AI into the development of AI models aimed at assisting in the classification of the presence or absence of residual thyroid tissue following surgical treatments. Commencing from the foundational concept of employing SPECT images in conjunction with noise filtering techniques as input data for a Resnet-18 model,⁶ the objective is to classify the presence or absence of residual thyroid tissue. However, when tested with our dataset, this method no longer maintains its accuracy, as our dataset contains more noise, resembling real-world scenarios encountered by medical professionals. This is demonstrated in the paper "Deep Learning Method Using SPECT Images to Diagnose Remaining Thyroid Tissue Post-Thyroidectomy".¹

Following these experiments, when working with a larger dataset and a more substantial model, notable enhancements in classification performance were

observed. This substantiates the viability of incorporating AI into the diagnostic process, suggesting that the application of a sufficiently expansive model along with an extensive dataset could potentially facilitate the classification of residual and non-residual thyroid tissue solely based on SPECT images.¹ With these promising results, the deep learning method for detecting the location of residual thyroid tissue post-surgery has been implemented, extending beyond mere classification into residual or non-residual categories.⁷ Based on existing research results, we want to improve and develop an end-to-end flow to support doctors with high accuracy in patient screening.

3 | DATASET DIVISIONS AND DIAGNOSIS PERFORMANCE METRICS

The SPECT image dataset was meticulously collected and prepared to ensure its suitability for accurate assessment of residual thyroid tissues post-thyroidectomy. The following steps outline the data split and preprocessing procedures.

3.1 | Data split

To facilitate rigorous development and evaluation of the RR-HCL SVM framework, the 430-sample dataset was deliberately partitioned into two distinct subsets: a training set and a test set. The training set, accounting for 80% of the data ($n = 340$), comprised equally balanced sub-populations of residual (positive) and nonresidual (negative) cases ($n = 165$ each). Similarly, the test set, encompassing 20% of the data ($n = 100$), contained an equal distribution of positive and negative cases ($n = 50$ each). This meticulously chosen data split ensured comprehensive representation of the entire sample population in both the training and testing phases, thereby enabling robust assessment of the RR-HCL-SVM framework's performance.

The models will be evaluated on the training set using the leave-one-out technique 3.3. Subsequently, the highest-scoring model will be selected and re-evaluated on the test set to provide the final results.

3.2 | Additional data

To enhance the credibility of the framework's performance evaluation, diagnostic results from multiple

experienced doctors were collected from 192 cases for comparison and evaluation.

3.3 | Diagnosis performance metrics

The robust evaluation of the proposed RR-HCL SVM framework for the assessment of residual thyroid tissues post-thyroidectomy in SPECT images is imperative to gauge its efficacy and reliability. To comprehensively assess the framework's diagnostic performance, we employ a set of well-established evaluation metrics. Furthermore, given the uniqueness of our dataset and the sensitivity of the task, we adopt the leave-one-out cross-validation (LOOCV) method for rigorous evaluation.

- *F1 Score*: The F1 score, which harmonizes precision and recall, is a pivotal metric in evaluating the model's ability to correctly classify residual thyroid tissues while minimizing false positives and false negatives. It is computed as:

$$F1 = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}}$$

- *Specificity*: Specificity quantifies the model's capacity to accurately identify non-residual regions or true negatives among all actual negatives. It is defined as:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

- *Sensitivity*: Sensitivity, also known as recall or true positive rate, assesses the model's ability to detect true positives among all actual positives, including residual thyroid tissues. It is calculated as:

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

To ensure a robust evaluation of the RR-HCL SVM framework, we employ the LOOCV method, a variant of cross-validation suitable for small datasets like ours. LOOCV systematically evaluates the model's performance by training it on all data points except one, which is held out for testing. This process iterates for each data point, providing a comprehensive assessment of the model's generalization capabilities.

These metrics collectively enable us to ascertain the framework's diagnostic accuracy, robustness, and real-world applicability.

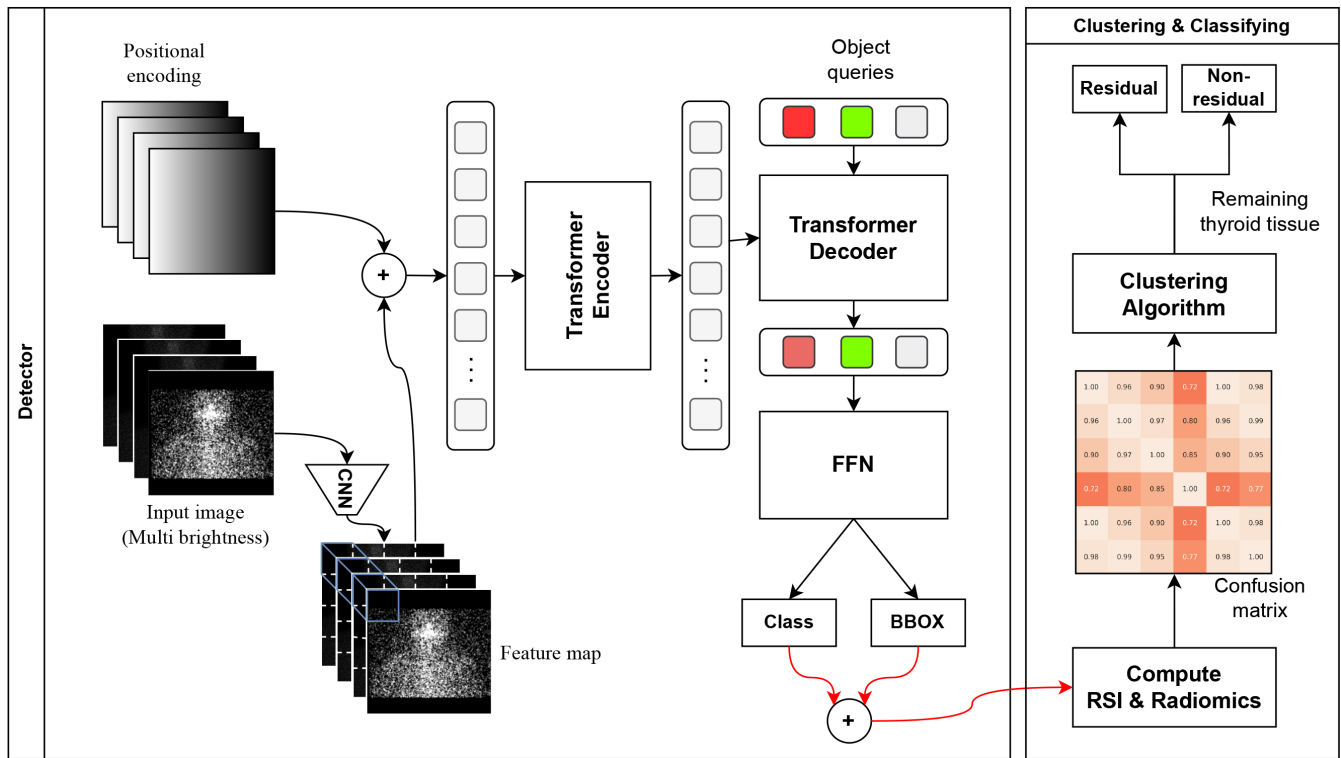


FIGURE 1 Architecture of RR-HCL-SVM a two-stage model for detecting and assessing residual thyroid tissues. The procedure entails using the DETR model to detect and recognize the thyroid and shoulder regions. After establishing these bounding boxes (where the red boxes indicate the model's predictions, and the green boxes indicate the ground truth), the RSI and radiomics measurements can be computed for the purpose of diagnosing any residual thyroid tissue.

4 | METHODOLOGY

4.1 | Overview proposed RR-HCL-SVM framework for assessing remaining thyroid tissue

The overall architecture of our solution is described in Figure 1. Initially, the raw SPECT image undergoes pre-processing, including contrast adjustment and the removal of the body part, as discussed in Section 5.4, before being fed into the model. Essentially, the 2-stage model comprises two main parts: detection and classification. The detection one is responsible for identifying the position of the neck and background (in this case, the shoulders) in the SPECT image. Determining both the neck and background positions is the initial step to calculate the RSI (Radiological Severity Index) and radiomics features.⁸ In the experimental section, we will demonstrate that these metrics exhibit high classification accuracy when used for diagnosing whether patients have excess armor or not.

We employ the DETR model⁹ for the detection stage due to its ability to learn the spatial relationships between objects in the image, a capability that traditional

CNN-based models lack. This capability of DETR is achieved through the attention mechanism within the Transformer encoders.

After identifying the neck and background regions of interest (ROIs), these bounding boxes are utilized to compute the RSI and radiomics features, along with the normalization of these new features. In an effort to reduce the dimensionality of features used for the classification model, we perform clustering on the RSI and radiomics features.

Clustering¹⁰ involves grouping similar data points, thereby simplifying complex datasets. This process reduces dimensionality, uncovers hidden patterns, and enhances machine learning model performance. By transforming data into meaningful representations, clustering not only maintains accuracy but also enhances the model's precision and optimization speed. We will substantiate this through a comparison of the classification model's effectiveness when using combinations of radiomics features with different clustering algorithms or without performing clustering.

Finally, in the classifier stage, these features are then passed through a straightforward classifier to make a binary diagnosis of whether the patient has excess armor

TABLE 1 Proposed model architecture layers and corresponding input–output shape. Here, N_{query} is a hyper-parameter, set to 75 for this study.

Layer	Input	Output
<i>Image Encoding</i>		
Brightness augmentation	$B \times 1 \times 256 \times 256$	$B \times 4 \times 256 \times 256$
ResNet50 backbone	$B \times 4 \times 256 \times 256$	$B \times 2048 \times 8 \times 8$
Conv2d 1×1	$B \times 2048 \times 8 \times 8$	$B \times 256 \times 8 \times 8$
Sin Positional Embedding	$B \times 256 \times 8 \times 8$	$B \times 256 \times 8 \times 8$
Query Embedding	$B \times N_{query}$	$B \times 256 \times N_{query}$
<i>Vanilla Transformer</i>		
Encoder	$B \times 256 \times 8 \times 8$	$B \times 256 \times 8 \times 8$
Decoder	$(B \times 256 \times 8 \times 8, B \times 256 \times N_{query})$	$B \times N_{query} \times 256$
<i>Class head</i>		
Fully connected	$B \times N_{query} \times 256$	$B \times N_{query} \times 3$
<i>Boxhead</i>		
Fully connected	$B \times N_{query} \times 256$	$B \times N_{query} \times 3$
Fully connected	$B \times N_{query} \times 256$	$B \times N_{query} \times 256$
Fully connected	$B \times N_{query} \times 256$	$B \times N_{query} \times 4$

or not, a binary classification task. We will also compare the classification effectiveness with two widely used machine learning models, SVM (Support Vector Machine) and Logistic Regression.

4.2 | DETR for residual detection

DETR (DEtection TRansformers) is a cutting-edge model in the realm of computer vision, renowned for its capacity to capture intricate spatial relationships within images. Leveraging a transformer-based architecture,¹¹ originally designed for natural language processing, DETR excels at object detection tasks by effectively identifying object classes and their precise locations. Unlike traditional methods, DETR operates end-to-end, eliminating the need for conventional object proposal mechanisms. Its self-attention mechanisms enable it to capture contextual information, making it highly adept at handling complex spatial arrangements of objects within images. This spatial awareness has made DETR a groundbreaking advancement in the field of computer vision, offering a more streamlined and effective approach to object detection. The study conducted by the previous study⁷ demonstrates that the DETR model yields outstanding results compared to other CNN-based detectors such as Faster-RCNN and YoloV7. In their research, the DETR model achieved a remarkable mAP@0.5 score of 0.43 on a dataset with characteristics similar to the dataset used in this study. Table 1 presents the sequential output shapes for each layer within our model.

4.3 | Integrating RSI and radiomics feature

In this study, we will utilize Radiological Severity Index (RSI) in conjunction with radiomics features calculated based on the detection results. Regarding RSI, this metric was previously mentioned in our study,⁷ and their research demonstrated that RSI is an effective categorical index, achieving an F1 score of 0.91. The formula for calculating RSI is as follows:

$$U(R) = \sum_{(x,y) \in E} I(x,y); RSI(B_T, B_S) = \frac{U(E_T)}{U(E_S)} \quad (1)$$

where: $I(x,y)$ is pixel value at position (x,y) ; E_T, E_S is the ellipse introspected of thyroid bounding box B_T and shoulder bounding box B_S ; $U(R)$ is function calculating uptake in the region R .

Besides, Radiomics feature extraction is a computational technique applied to ROIs within medical images, particularly in the field of radiology.¹² It involves the quantitative analysis of image data to extract a wide array of high-dimensional features, such as texture (GLCM, GLRLM, GLSZM), shape (Area, Perimeter, Elongation, etc), and intensity statistics (Mean, Standard deviation, etc) from the selected ROI. These features encapsulate intricate information about the underlying tissue characteristics, which may not be discernible to the naked eye. We use a total of 84 features including 83 radiomics features and final RSI one. All used radiomics features are available in Appendix A.

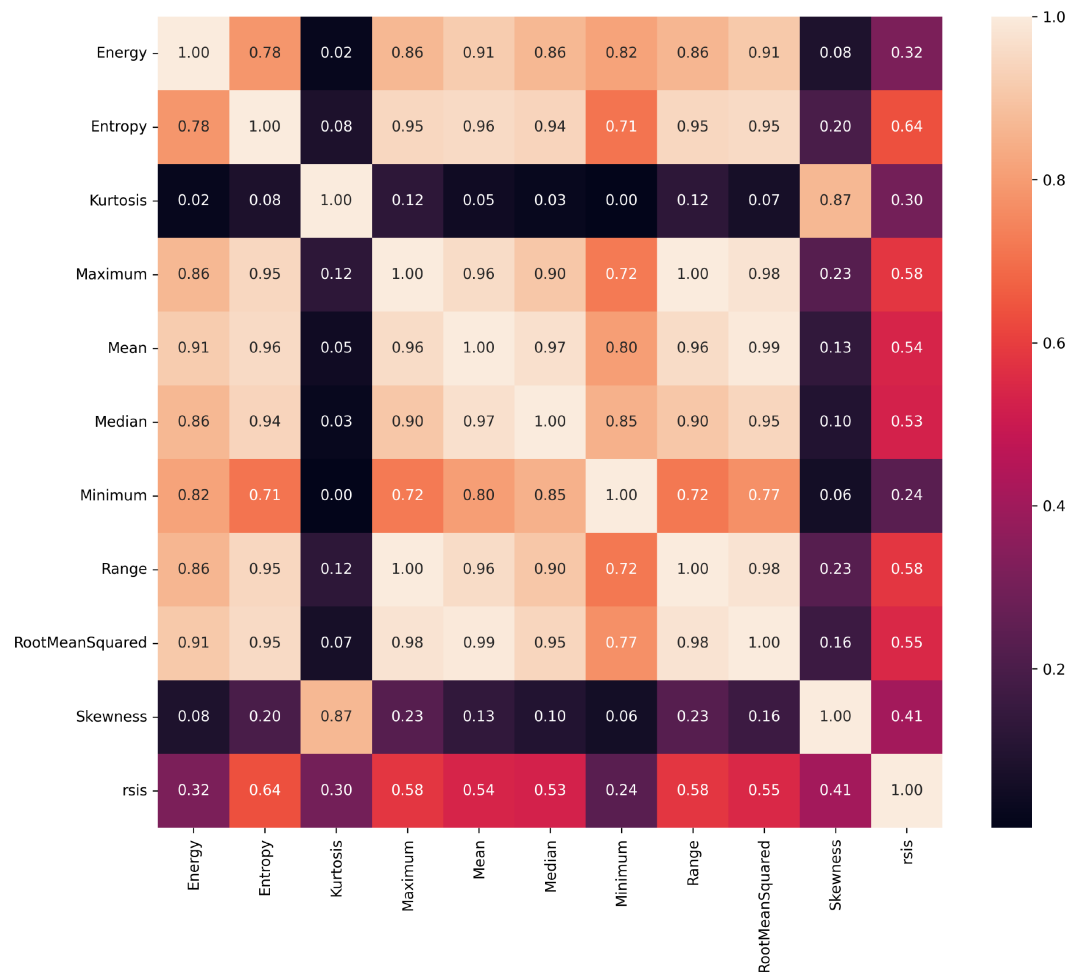


FIGURE 2 The partial correlation matrix spotlights relationships that might guide our feature selection quest. Tune in to our features used to build full version of matrix in Appendix A.

In this research paper, we will demonstrate that the inclusion of radiomics features calculated over the Region of Interest (ROI) in the neck region enhances the accuracy of the classification model.

4.4 | Feature clustering

Radiomics features, which provide valuable insights into tissue characteristics, are often extracted from medical images to aid in diagnosis and assessment. However, the sheer volume of these features, coupled with their inter-related nature, can lead to multicollinearity and, consequently, suboptimal model performance. Highly correlated features not only contribute to increased computational complexity but can also introduce bias and overfitting, compromising the robustness of our analysis. This approach groups highly correlated features into clusters, reducing dimensionality while preserving the critical information inherent in the data. Finally, we choose the

mean value of all features in each group as only new feature for group and ignore others. This not only streamlines the computational burden but also enhances the interpretability of the model, ensuring that the selected features are truly representative of the underlying tissue characteristics.

Figure 2 shows the confusion matrix of partial features using Pearson Correlation.¹³ It is obviously many highly correlated features, thus employing clustering algorithm in this problem is really ideal.

4.4.1 | Building connectivity graph

In our analysis, we employ a connectivity graph to uncover intricate relationships among extracted radiomics features. This graph serves as a powerful tool to reveal feature interdependencies. We compute Pearson correlation coefficients¹³ $\rho(i,j)$ between feature pairs using the formula:

$$\rho(i,j) = \frac{\sum ((X_i - \bar{X})(X_j - \bar{X}))}{\sqrt{\sum (X_i - \bar{X})^2} \cdot \sqrt{\sum (X_j - \bar{X})^2}} \quad (2)$$

Here, X_i, X_j are feature values, and $\bar{X}$ is the mean. The coefficient ranges from -1 (perfect negative correlation) to 1 (perfect positive correlation), with 0 indicating no correlation. We set a threshold of 0.9 for the absolute value of $\rho(i,j)$. Features with $|\rho(i,j)|$ exceeding this threshold indicate strong connections, forming the basis for constructing our connectivity graph. This graph guides dimensionality reduction through feature clustering, facilitating the discovery of underlying feature patterns. Thus we formed connectivity graph $G \in \{0,1\}^{n \times n}$ in where n is number of features. G is directly fed as input of our following clustering algorithms.

4.4.2 | Spectral clustering

Spectral clustering¹⁴ is a powerful algorithm for partitioning data points into clusters based on their similarity, leveraging the connectivity graph provided as input. This method takes advantage of the graph's structural information and spectral properties to identify meaningful clusters. The steps of spectral clustering are as follows:

1. **Graph Laplacian:** Given the pre-existing connectivity graph, the first step involves constructing the graph Laplacian matrix, typically defined as the degree matrix (D) minus the adjacency matrix (A), which characterizes the graph's connectivity.

$$L = D - A$$

2. **Eigenvalue Decomposition:** Proceed to compute the eigenvectors and eigenvalues of the Laplacian matrix L . These eigenvectors capture the graph's intrinsic structure and spectral information.

$$L \cdot v = \lambda \cdot v$$

3. **Dimensionality Reduction:** Select a subset of eigenvectors corresponding to the smallest eigenvalues. These eigenvectors represent the data points in a reduced-dimensional space while preserving the essential cluster structure.
4. **K-Means Clustering:** Apply a clustering algorithm, such as K-means, to the reduced-dimensional space formed by the selected eigenvectors. This step assigns data points to clusters based on their transformed representations.

Spectral clustering's strength lies in its ability to unveil complex cluster patterns within data when provided with a connectivity graph. By harnessing the spectral properties of the graph Laplacian, it offers a powerful method for cluster discovery in various applications, such as image segmentation, community detection in networks, and pattern recognition. Spectral clustering serves as making it a versatile and effective technique in unsupervised machine learning.

4.4.3 | Hierarchical clustering

Hierarchical clustering¹⁵ is a data clustering technique used to group similar data points into clusters in a hierarchical, tree-like structure called a dendrogram. The key behind hierarchical clustering is to iteratively merge or split clusters to reveal the underlying structure of the data. The steps of hierarchical clustering are as follows:

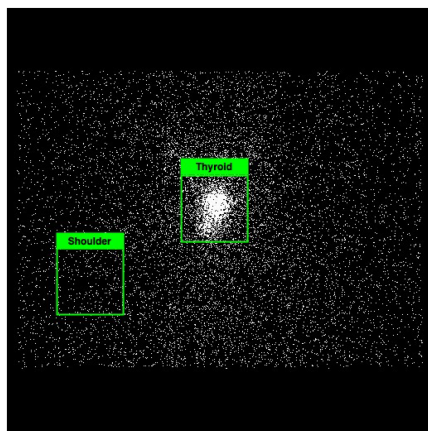
1. **Initialization:** Start by considering each data point as its own cluster, creating as many initial clusters as there are data points.
2. **Agglomerative Approach:** In agglomerative hierarchical clustering, the algorithm begins to merge the most similar clusters based on a chosen linkage criterion. Common linkage methods include single linkage (merging the closest data points or clusters), complete linkage (merging clusters with the closest Maximum pairwise distance), and Average linkage (merging clusters with the closest Average pairwise distance).
3. **Dendrogram Construction:** As clusters are merged, a dendrogram is constructed, which is a hierarchical tree structure that visually represents the merging process. The dendrogram can be used to explore clusters at different levels of granularity.
4. **Dendrogram Cutting:** Clusters can be selected from the dendrogram at different heights, allowing for flexibility in determining the number and size of clusters.

Hierarchical clustering can be used in various fields, including biology, image analysis, and social sciences, to uncover hidden structures and patterns within data.

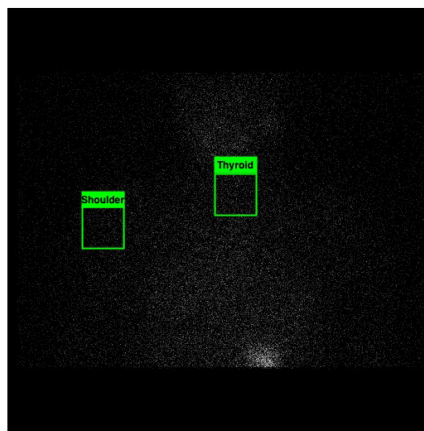
5 | DATA ACQUISITION AND PREPROCESSING

5.1 | Data acquisition

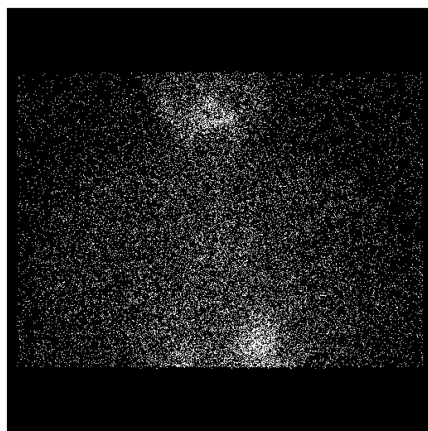
The dataset comprises two main types of SPECT images—anterior whole-body scans (ANT-WBS) with dimensions of 1024×256 pixels and anterior static



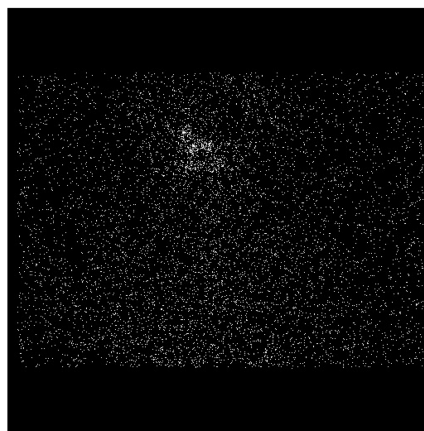
(A) Body contours appear blurry, but the shielding's radiation capture is more pronounced.



(B) Body contours are more distinct, but the uptake of thyroid tissue is less clear.



(C) Body contours are more distinguishable, but there are also other organs that capture ^{131}I radiation.



(D) High level of noise image, the cervical and shoulder region are confused.

FIGURE 3 Description of SPECT images from our dataset: The two images above depict cases with residual thyroid tissue, while the two images below showcase with complete removal.

images with dimensions of 512×512 pixels. These images were acquired using state-of-the-art SPECT imaging equipment at 108 Military Central Hospital. Imaging parameters, including gamma camera model, energy window settings, and acquisition time, were carefully calibrated to obtain high-quality scans.

5.2 | Annotation

Expert radiologists meticulously annotated each image in the dataset. Annotations were made to identify ROI, specifically the residual thyroid tissue and the shoulder, which served as the background region. In cases where patients had residual thyroid tissues, both regions were enclosed within a common rectangular bounding box as

Figure 3. This annotation process was a critical step to provide ground truth labels for subsequent analysis.

5.3 | Challenges

Despite careful acquisition and preprocessing, the dataset presents specific challenges as Figure 3:

1. *Blurry Boundaries*: The boundaries between the shoulder and head regions often exhibit blurriness, making accurate detection of the background region challenging.
2. *Poor contrast*: Locating the radioactivity associated with the thyroid tissue can be intricate in certain cases.

5.4 | Image preprocessing

Prior to model training and testing, the SPECT images underwent a series of preprocessing steps. These included cropping of irrelevant body parts from the ANT-WBS scans, retaining only the head-neck region thus final images are 512×512 . Additionally, histogram equalization was applied to enhance image contrast. These preprocessing steps were essential for optimizing image quality and consistency across the dataset while focusing on the relevant anatomical Area.

6 | EXPERIMENT

6.1 | DETR settings for residual thyroid tissue detection

We employ the DETR model for the stage of detecting excess armor and shoulder positions. Instead of training the DETR model from scratch, we leverage a pretrained model made publicly available by Phu et al.⁷ The DETR model used in this study was originally trained on SPECT image data, allowing us to benefit from transfer learning. The configuration for the model, as described in Table 2, outlines the roles of individual parameters.

6.2 | Optimized model parameter ranges

The model optimization process encompasses the fine-tuning of both the model's internal parameters and hyperparameters using Adam optimizer¹⁶ with 100 epochs. While the model parameter updates can be automated, defining a specific search range is essential for hyperparameters. The RR-HCL model consists of two stages: the DETR detection model and the classification model. The DETR model will directly utilize a pretrained model from the research by Phu et al.,⁷ and its weights will be frozen, eliminating the need to set hyperparameter ranges for DETR. The remaining optimization tasks will be focused on the classification model and the clustering part. Table 3 illustrates the search ranges for all parameters used in the optimization of the RR-HCL model, encompassing parameters for the classification model and clustering.

In the case of the clustering model, before performing clustering, a connectivity graph is constructed based on the correlation matrix between pairs of features, computed using Equation 2. In this graph, each vertex represents a feature, and whether two vertices are connected or not depends on whether $|\rho(i,j)| > \rho - \text{threshold}$ or not. Therefore, the $\rho - \text{threshold}$ determines how the

TABLE 2 DETR configuration for detection task.

Backbone	Brightness	ϵ	C	Class
ResNet50	4	0.3	4	2

feature groups are distributed and serves as an important hyperparameter. As the graph G may not be fully connected, for connected components with fewer vertices than a certain threshold, which we specifically refer to as the *group-cut-off*, clustering will not be necessary. Figure 4 shows the desired connectivity G .

The algorithm employed for clustering is also a hyperparameter in our study. We utilize two clustering algorithms known for their effectiveness in numerous research studies: Spectral and Hierarchical clustering. For each clustering algorithm, the target number of clusters is another hyperparameter, which we denote as *Clusters*.

Finally, there are parameters specific to the classification models, which include Logistic Regression and SVM. For SVM, the hyperparameters include the degree of the polynomial kernel, denoted as *Degree*, and the common parameters shared by both classifiers: the regularization term C and the penalty method *Penalty*, which can be either L_1 or L_2 . The model configuration yielding the best performance during Grid Search¹⁷ will undergo a final leave-one-out evaluation to determine its suitability for further analysis.

6.3 | Results

6.3.1 | The effect of radiomics features

In this section, we conducted experiments to validate whether the inclusion of radiomics features enhances the accuracy of the model for diagnosing excess armor presence in the thyroid. The test data was evaluated using SVM and Logistic Regression models, both with and without the integration of radiomics features. The experimental results are presented in Table 4. This result table demonstrates that the incorporation of radiomics features leads to significant improvements in specificity, sensitivity, and F1 score compared to models using only RSI as a feature. Specifically, sensitivity increased by 4.2%, specificity increased by 2.2%, and F1 score increased by 3.22%. Notably, the optimal penalty term also increased from 10 to 1000 when radiomics features were integrated.

In our analysis, we also compared the performance of models with and without radiomics features call: RR-Logistic and RSI-Logistic. Figure 5 illustrates the ROC curve associated with each respective model. Notably, the

TABLE 3 Optimization search domain of clustering and classification models with different parameters.

Model	Clustering algorithm	Degree	Penalty	group-cut-off	C	ρ -threshold	Clusters
SVM RSI only	-	[1..4]	$[L_1, L_2]$	-	[0.1,1,10,100]	-	-
Logistic RSI only	-	-	$[L_1, L_2]$	-	[0.1,1,10,100,1000]	-	-
SVM RR (no clustering)	-	[1..7]	$[L_1, L_2]$	-	[0.1,1,10,100]	-	-
Logistic RR (no clustering)	-	-	$[L_1, L_2]$	-	[0.1,1,10,100,1000]	-	-
SVM RR	[Spectral, Hierarchical]	[1..5]	$[L_1, L_2]$	[3,5,8]	[0.1,1,10,100]	[0.7 – 0.95]	[2,3,5]
Logistic RR	[Spectral, Hierarchical]	-	$[L_1, L_2]$	[3,5,8]	[0.1,1,10,100,1000]	[0.7 – 0.95]	[2,3,5]

Note: “RR” mean both RSI and radiomics feature are used as features.

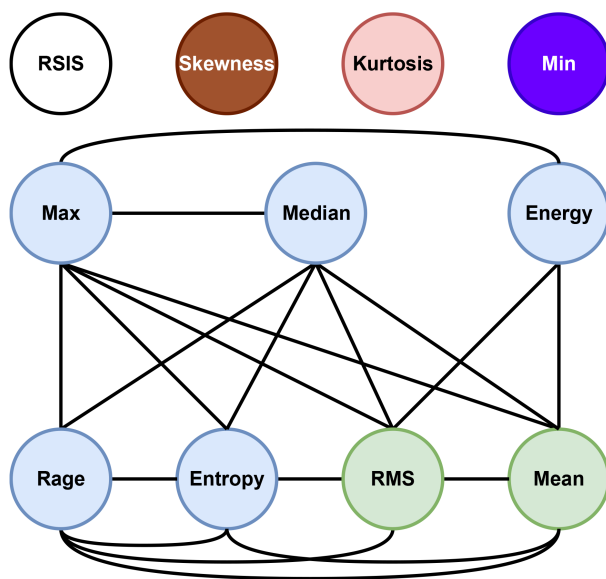


FIGURE 4 A derived connectivity graph was constructed based on a correlation matrix Figure 2 with a correlation threshold of 0.9, the number of clusters for each connected component was set to three, in which the vertices that belong to the same connected component are represented by the same color.

Area under the curve (AUC) of the RR-Logistic model, which incorporates radiomics features, achieved an AUC of 0.851. In contrast, when using only the RSI without radiomics features, the AUC dropped to 0.785. This finding underscores the importance of radiomics features in improving the discriminatory power of the model, the same as the conclusion on sensitivity and F1 scores.

6.3.2 | The effect of clustering

We perform a series of experiments, with different features, clustering algorithms, and classification

algorithms. For each clustering configuration, we follow the entire pipeline of the RR-HCL-SVM framework, which includes feature extraction, feature clustering, and classification using the classification model. The test results of our experiments are summarized in Table 4 along with the best configuration of corresponded models.

Table 4 demonstrates that the SVM model, when utilizing the combination of radiomics features and RSI along with Hierarchical clustering, achieves the best results with an F1 score of 0.97. Compared to the model without clustering, sensitivity has improved by 3.22%, rising from 0.93 to 0.96.

Based on the number of features after clustering, we calculate the dimensionality reduction with respect to the initial number of features, and the number of features after clustering. Dimensional reduction is computed using the following formula:

$$D - \text{reduction} = \frac{\text{old_n_features} - \text{new_n_features}}{\text{old_n_features}} (\%)$$

where: old_n_features is the initial number of features (84 in this context); new_n_features is the number of features after clustering, which varies depending on the specific clustering algorithm used.

The results in Table 4 demonstrate a substantial reduction in the number of features (over 80% reduction is achievable) without compromising diagnostic accuracy significantly. Particularly, when employing a Hierarchical approach in conjunction with a simple classification model such as Logistic Regression, we only need to utilize 10% of the original set of features to achieve comparable outcomes.

This indicates that grouping features together not only reduces feature dimensionality, which accelerates the optimization process but also enhances the model's accuracy. The use of different clustering algorithms leads

TABLE 4 Classification results of different models with best parameters.

Model	Clustering algorithm	Specificity	Sensitivity	F1	Group-cut-off	Degree	<i>n</i> clusters	ρ -threshold	D-reduction (%)	C
SVM RSI only	-	0.94	0.91	0.93	-	1	-	-	-	10
Logistic RSI only	-	0.94	0.91	0.93	-	-	-	-	-	10
SVM RR (no clustering)	-	0.98	0.93	0.96	-	1	-	-	-	100
Logistic RR (no clustering)	-	0.98	0.93	0.96	-	-	-	-	-	1000
SVM RR	Spectral	0.98	0.93	0.96	5	-	2	0.9	81	1
	Hierarchical	0.98	0.96	0.97	5	1	3	0.9	78	100
Logistic RR	Spectral	0.98	0.93	0.96	5	1	2	0.9	81	10
	Hierarchical	0.98	0.93	0.96	8	50	2	0.7	90	1

Note: “RR” mean both RSI and radiomics feature are used as features. Bold values denote the best values.

to varying results, but overall, compared to using a model that employs all features, the accuracy is not compromised. Therefore, clustering proves to be an effective method for dimensional reduction.

There is no significant change in the specificity among different classification models with various clustering algorithms. However, there is a noticeable difference in sensitivity. In the medical field, particularly in thyroid cancer diagnosis, sensitivity is a crucial metric as it ensures that no patients are missed. The improvement in sensitivity due to clustering is a strong signal indicating the potential practicality of the proposed model.

Figure 5 illustrates that although using clustering on radiomics-combined features leads to improved F1 scores, the clustering algorithm coupled with the Logistic Regression model results in a lower AUC of approximately 0.73 compared to not using clustering, which yields an AUC of 0.851. In the case of the SVM model, there is a notable improvement with an AUC of approximately 0.99. This underscores the significance of selecting the classification model, as it profoundly influences the model's classification capabilities. Hence, we opt to utilize a first-degree SVM model combined with Hierarchical clustering as the proposed model.

6.3.3 | Comparison with gold standard

Following the assessment of these models' performance, our research team proceeded with additional experiments for post-treatment diagnosis. We combined the outcomes of our best-performing model with RSI, radiomics, and clustering results. Table 5 is a comparative result displaying diagnostic outcomes in comparison to the diagnoses made by two medical professionals on the identical dataset, comprising 192 post-treatment patients. The benchmark for assessment relies on the diagnosis made by a renowned specialist in the field, marked with gold standard labels.

Analyzing the table that juxtaposes the diagnostic outcomes regarding thyroid gland radiation emission, we observe that the model exhibits Precision and Recall values that are better than those achieved by the doctors' diagnoses. Furthermore, the F1 score remains consistent for both residual and nonresidual cases.

7 | DISCUSSION

Thyroid cancer is an emerging area that has ample scope for applying AI in traditional diagnostics. While fully automating the diagnosis and treatment process is a long-term goal, this study focused on improving the accuracy

of diagnosing successful tissue deletion in the post-operative treatment stage. Improving accuracy and developing new problem-solving models can enhance productivity, reduce the workload of doctors, and minimize the time and financial burden on thyroid cancer patients.

The evaluation and identification of surplus bone armor in the application of AI remain relatively nascent, with scant research dedicated to this class of problems. The task of assessing surplus bone armor can be addressed as a typical binary classification problem, exemplified by conventional classification models such as ResNet-18⁶ (F1 of 0.88) or ResNet-50¹ (F1 of 0.86). However, the extraction of features and direct assessment of

noisy SPECT images is not a practical choice in reality. The quality of SPECT images depends on various factors, including the detector, the patient's medication administration duration, health condition, and the prevalence of SPECT images with unclear body contours. Consequently, employing GrabCut⁶ yields suboptimal results. Additionally, in SPECT images, not only is the surplus bone armor capturing radiation evident, but salivary glands exhibit a similar pattern. Therefore, when the model cannot accurately determine the cervical position due to blurred body contours, it may erroneously focus on these glands, leading to inaccurate conclusions.

Instead of directly evaluating the original SPECT images in this study, we have demonstrated that feature extraction statistics on the Region of Interest (RSI) in the neck (as a result of the detection model) along with the RSI index indicate a robust correlation between the cervical region and the background radiation, proving more resilient to noise. The average F1 score achieved is 0.97 on the same dataset as in⁶ and.¹ These features are also aligned with the diagnostic process of physicians, as in reality, the presence of surplus bone armor in patients depends on the radiation correlation and the absorptive characteristics of the cervical region.

Nevertheless, this article still harbors certain limitations that could be addressed for improvement in future research. The following idea suggests potential future improvements:

1. The model employs a fixed amount of adjustment for input images with varying brightness levels. As a result, we can build a model that can predict the suitable brightness levels for each image.
2. Carefully, turning hyper parameter consisting of number query, number of encoder, decoder class_cost within DETR architecture.

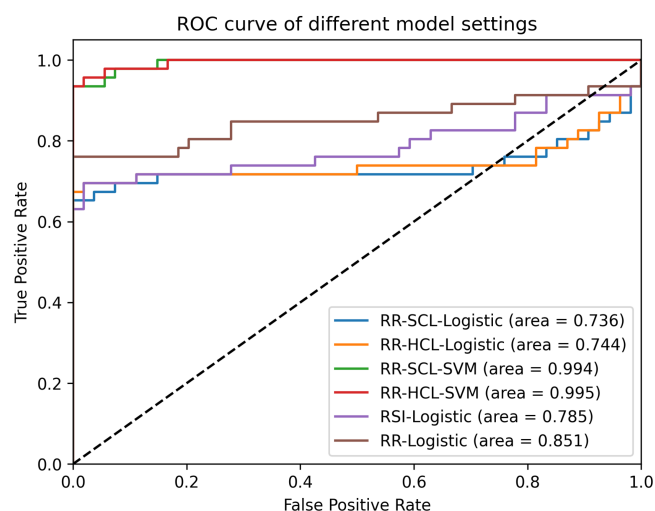


FIGURE 5 ROC curve on test set with different models. Where “RR” means model both RSI and Radiomics features; “SCL” means model uses spectral clustering; and “HCL” means Hierarchical clustering.

TABLE 5 Compare diagnosis results of proposed model with hierarchical clustering to the actual predictive results of the doctors.

Name	Class	Precision	Recall	F1
Our proposal	Nonresidual	0.96	0.82	0.89
	Residual	0.8	0.96	0.88
Guo et al. ⁶	Nonresidual	0.82	0.82	0.81
	Residual	0.79	0.77	0.78
Our previous study ¹	Nonresidual	0.86	0.89	0.87
	Residual	0.88	0.85	0.86
1st Doctor	Nonresidual	0.96	0.61	0.75
	Residual	0.74	0.98	0.84
2nd Doctor	Nonresidual	0.90	0.67	0.76
	Residual	0.76	0.93	0.84

Note: Bold values denote best values.

3. During tissue removal treatment, doctors often rely on their experience to determine the appropriate dosage, which can result in either an overdose or an insufficient dosage to remove all excess thyroid tissue in one attempt. Therefore, it is crucial to diagnose the individual tissue removal dose, which is sufficient to remove excess thyroid tissue, by utilizing scintigraphy, blood tests, size, and segmentation results.

8 | CONCLUSION

In conclusion, our research culminates in the development of an advanced model for the detection and assessment of residual thyroid tissues, yielding remarkable results. We propose a two-stage deep learning model, which not only enhances the accuracy of thyroid tissue detection but also provides a comprehensive assessment of the residual tissues.

Leveraging both the structural information from RSI and the quantitative radiomics features, and applying clustering for dimensional reduction allowed us to capture a more comprehensive view of thyroid tissues, resulting in a more effective detection and assessment process.

Our best-performing model, termed RR-HCL SVM, achieves an impressive AUC of 0.995, showcasing its superior discriminative power. With a specificity of 0.98, sensitivity of 0.96, and an F1 score of 0.91, our model exhibits exceptional performance in distinguishing between thyroid tissues, demonstrating its clinical relevance.

This research paves the way for more accurate and efficient thyroid tissue assessment, ultimately benefiting patients and healthcare practitioners.

AUTHOR CONTRIBUTIONS

Minh Lai Phu provided the general idea for this research work, designed the model, analyzed the data, and wrote the manuscript. Thanh Vinh Pham designed the model, carried out the implementation, and wrote the manuscript. Thuc Pham Duc carried out the implementation. Trung Nguyen Thanh, and Long Tran Quoc shared critical analysis and suggestions for the experiment and manuscript revision. Duc Chu Minh, Ha Le Ngoc, Son Mai Hong, Phuong Nguyen Thi, Nhung Nguyen Thi, and Khanh Le Quoc analyzed the data, reviewed and edited the manuscript. Thuan Duc Nguyen, Ha Nguyen Thai reviewed and edited the manuscript. Thanh Nguyen Chi provided the general idea for this research work, analyzed the data, and shared critical analysis and suggestions for the experiment and manuscript revision.

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CONFLICT OF INTEREST STATEMENT

The authors declare that there is no conflict of interest regarding the publication of this article.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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APPENDIX A: RADIOMICS FEATURES

Below is a detailed list of all the radiomics features utilized in this research:

10Percentile, 90Percentile, Energy, Entropy, Interquartile Range, Kurtosis, Maximum, Mean Absolute Deviation, Mean, Median, Minimum, Range, Robust Mean Absolute Deviation, Root Mean Squared, Skewness, Total Energy, Uniformity, Variance, Elongation, Flatness, Least Axis Length, Major Axis Length, Maximum 2D Diameter Column, Maximum 2D Diameter Row, Maximum 2D Diameter Slice, Maximum 3D Diameter, Mesh Volume, Minor Axis Length, Sphericity, Surface Area, Surface Volume Ratio, Voxel Volume, Auto correlation, Cluster Prominence, Cluster Shade, Cluster Tendency, Contrast, Correlation, Difference Average, Difference Entropy, Difference Variance, Id, Idm, Idmn, Idn, Imc1, Imc2, Inverse Variance, Joint Average, Joint Energy, Joint Entropy, MCC, Maximum Probability, Sum Average, Sum Entropy, Sum Squares, Gray Level Non Uniformity, Gray Level Non Uniformity Normalized, Gray Level Variance, High Gray Level Run Emphasis, Long Run Emphasis, Long Run High Gray Level Emphasis, Long Run Low Gray Level Emphasis, Low Gray Level Run Emphasis, Run Entropy, Run Length Non Uniformity, Run Length Non Uniformity Normalized, Run Percentage, Run Variance, Short Run Emphasis, Short Run High Gray Level Emphasis, Short Run Low Gray Level Emphasis, High Gray Level Zone Emphasis, Large Area Emphasis, Large Area High Gray Level Emphasis, Large Area Low Gray Level Emphasis, Low Gray Level Zone Emphasis, Size Zone Non Uniformity, Size Zone Non Uniformity Normalized, Small Area Emphasis, Small Area High Gray Level Emphasis, Small Area Low Gray Level Emphasis, Zone Entropy, Zone Percentage, Zone Variance.